Do age and transportation habits predict communication apprehension?

Tyler Johnston, Vanessa Kountakis, Shivya Paul, Jacob Van Vleet

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1 Introduction and Literature Review

This manuscript is our submission for the summer semester project for Psych 755 at UW-Madison and its course entitled *Tools for Large Scale Data Science*. In this project we demonstrate usage of the following tools: R (`tidyverse` and `tidymodels`), Python (`pandas`, `scikit-learn`, `matplotlib`, and `seaborn`), Quarto for reproducible reporting, and machine-learning pipelines for predictive modeling.

1.1 Literature Review

Communication apprehension, the fear or anxiety associated with real or anticipated communication with others, is a well-studied individual difference that has been linked to a range of social and behavioral outcomes. The Personal Report of Communication Apprehension (PRCA-24) is a widely used self-report measure of communication apprehension originally described by McCroskey in *An Introduction to Rhetorical Communication* (McCroskey 1982) and made available on the author's assessment page (McCroskey n.d.). Higher scores on the PRCA-24 indicate greater anxiety across interpersonal, group, meeting, and public speaking contexts.

Daily transportation choices and age may also relate to social experience. Ride-share services and public transportation require different levels of interaction with strangers, drivers, and fellow passengers. If communication apprehension shapes or is shaped by these everyday social exposures, we would expect transportation habits and age to be associated with PRCA scores. The present study tests these expectations in a single sample.

1.2 Research Questions and Hypotheses

We address four main research questions (RQs):

- **RQ1:** Do individuals who use ride share, public transportation, or both differ in communication apprehension from those who use neither mode of transportation?
  - H_0: There is no difference in mean communication apprehension between individuals who use at least one mode of transportation and those who use neither.

- H_a: Mean communication apprehension differs between individuals who use at least one mode of transportation and those who use neither.

- **RQ1.5:** Among individuals who use transportation, do communication apprehension levels differ by transportation mode?
  - H_0: There is no difference in mean communication apprehension across the four transportation mode groups.
  - H_a: At least one transportation mode group differs in mean communication apprehension.
- **RQ2:** Is age associated with communication apprehension?
  - H_0: Age is not associated with communication apprehension.
  - H_a: Age is associated with communication apprehension.
- **RQ3:** Does age moderate the relationship between transportation use and communication apprehension?
  - H_0: The association between overall transportation frequency and communication apprehension does not differ between younger (17–30) and older (31+) participants.
  - H_a: The association between overall transportation frequency and communication apprehension differs by age group.
- **RQ4:** Can machine-learning models predict communication apprehension from age and transportation habits more accurately than a baseline mean model, and which model performs best?
  - H_0: Machine-learning models do not improve prediction over the baseline mean model.
  - H_a: At least one machine-learning model achieves a lower RMSE than the baseline mean model.

2 Data and Methods

2.1 Data Preparation

2.1.1 Data Collection and Merging

Data were collected from participants who completed a survey assessing communication apprehension using the PRCA (McCroskey 1982) and transportation habits. The original data were stored in three separate files: `PRCAProlificExport_FileA.csv`, `PRCAProlificExport_FileB.csv`, and `PRCAQualtricsExport_FileC.csv`. The two Prolific files (`FileA` and `FileB`) contained participant demographic information including age, sex, country of residence, student status, and employment status. The Qualtrics file (`FileC`) contained survey responses including communication apprehension questions and transportation usage patterns.

The data cleaning process involved several steps. First, the two Prolific files were concatenated using Python's pandas library, resulting in a combined dataset with duplicate participant IDs removed (keeping the first occurrence). This yielded 252 unique participants. The combined Prolific data were then merged with the Qualtrics data by matching the `Q0` column in the Qualtrics file with the `Participant id` column in the Prolific file. Rows without matches (n = 31) were filtered out, resulting in a final dataset of 252 participants with complete demographic and survey data.

2.1.2 Variable Coding and Creation

After merging, the survey responses required coding to create numeric variables for analysis. Likert-scale responses to communication apprehension questions were reverse-coded where appropriate. Questions Q1, Q3, Q5, Q13, Q15, and Q18 were normally coded (Strongly disagree = 0, Somewhat disagree = 1, Neither agree nor disagree = 2, Somewhat agree = 3, Strongly agree = 4), while questions Q2, Q4, Q6, Q14, Q16, and Q17 were reverse-coded (Strongly disagree = 4, Somewhat disagree = 3, Neither agree nor disagree = 2, Somewhat agree = 1, Strongly agree = 0).

Reverse coding was necessary because some survey items were phrased in the opposite direction. For example, Q1 asks "I dislike participating in group discussions" (normal coded), where agreement indicates higher communication apprehension. In contrast, Q2 asks "Generally, I am comfortable while participating in group discussions" (reverse coded), where agreement indicates lower communication apprehension. Without reverse coding, these items would be scored in opposite directions, leading to inconsistent measurement. By reverse-coding Q2 (and similar items), we ensure that higher scores consistently represent higher communication apprehension across all items.

Transportation usage variables were created from survey questions Q26 (public transportation) and Q28 (ride-share). Binary variables indicated whether participants used each mode (Never = 0, any use = 1). Frequency variables were created with ordinal coding: Never = 0, 0-1 days a month = 1, 2-4 days a month = 2, 4-8 days a month = 3, 8 or more days a month = 4. Daily frequency variables were similarly coded for typical daily usage patterns.

A composite communication apprehension score was calculated by summing responses to the 12 communication apprehension questions (Q1-Q6 and Q13-Q18) and dividing by 5, resulting in a score representing average communication apprehension per question. Finally, a minimal dataset was created containing only the essential variables for analysis: participant ID, age, sex, transportation usage variables, and the communication apprehension score.

2.2 Descriptive Statistics

Table 1: Descriptive statistics for key continuous variables

Variable	M	SD	n
Age	39.24	13.41	252
Communication apprehension	1.41	0.94	252
Public transport frequency	2.08	1.44	252
Ride-share frequency	1.55	1.26	252

Table 1 summarizes the key continuous variables. Participants reported a mean communication apprehension score of 1.41 (SD = 0.94). Ride-share and public-transport frequency variables are ordinal measures coded from 0 (never) to 4 (most frequent use).

2.3 Overview of Analytic and Predictive Methods

This analysis used t-tests, regression, ANOVA, and correlation to test explanatory relationships and to answer the main inferential RQs. We also used machine learning to optimize prediction of communication apprehension and to answer our algorithmic RQ.

For the inferential questions, we used independent-samples t-tests to compare mean differences between transportation groups, one-way ANOVA to compare mean communication apprehension across categorical frequency groups, and a moderation model to test the interaction between age group and overall transportation frequency. All inferential models were fit in R and later translated to python, using Seaborn for visuals.

For the predictive question, we compared four models: a baseline that always predicted the mean communication apprehension, k-nearest neighbors (k = 5), a random forest (100 trees, mtry and minimum node size tuned via 5-fold cross-validation), and a feed-forward neural network (hidden units, activation, and penalty tuned via 5-fold cross-validation). Predictive performance was evaluated with root mean squared error (RMSE) on held-out data. The neural network used a 60/20/20 train/validation/test split, with the top model being trained on both the train and validation sets before predicting into test data; KNN and random forest used an 80/20 split. Models were fit and tuned with R `tidymodels`.

Table 2: Average communication apprehension by transportation group

	Transportation group	Mean communication apprehension per Question	N
0	Neither	2.111111	36
1	Ride share only	1.166667	15
2	Public transport only	1.592593	36
3	Both	1.234343	165

3 Results

3.1 RQ1: Do Transportation Habits Predict Communication Apprehension?

An independent-samples t-test found a significant difference in communication apprehension between participants who used ride share, public transportation, or both and participants who used neither mode, $t(250) = -5.08$, $p < .001$. Participants who used at least one mode of transportation reported communication apprehension scores that were, on average, 0.82 points lower per question (9.84 points lower across all questions) than participants who used neither mode (mean difference $= -0.82$, SE $= 0.16$).

3.1.1 RQ 1.5: Do individuals who use different modes of transportation differ in communication apprehension?

We first examined whether communication apprehension differed based on transportation mode usage. Participants were categorized into four groups based on their transportation habits: those who used neither ride-share nor public transportation, those who used only public transportation, those who used only ride-share, and those who used both modes.

As shown in Table 2, ride-share-only users had the lowest average communication apprehension scores (1.17), followed by those who used both modes (1.23), public transportation-only users (1.59), and those who used neither mode (2.11). This pattern again suggests that transportation mode is associated with communication apprehension, with ride-share users showing particularly low levels.

3.1.2 Transportation Frequency Effects

Beyond simple mode usage, we examined whether the frequency of transportation use was associated with communication apprehension. We conducted a linear regression analysis including both ride-share frequency and public transportation frequency as predictors of communication apprehension.

The analysis revealed that each one-category increase in ride-share frequency was associated with an average 0.23 point (per question) decrease in communication apprehension, accounting for public-transport frequency, $b = -0.23$, $SE = 0.05$, $t(249) = -4.28$, $p < .001$. In contrast, each one-category increase in public-transport frequency was associated with an estimated 0.05-point decrease in communication apprehension after accounting for ride-share frequency; however, this association was not statistically significant, $b = -0.05$, $SE = 0.05$, $t(249) = -1.00$, $p = .318$.

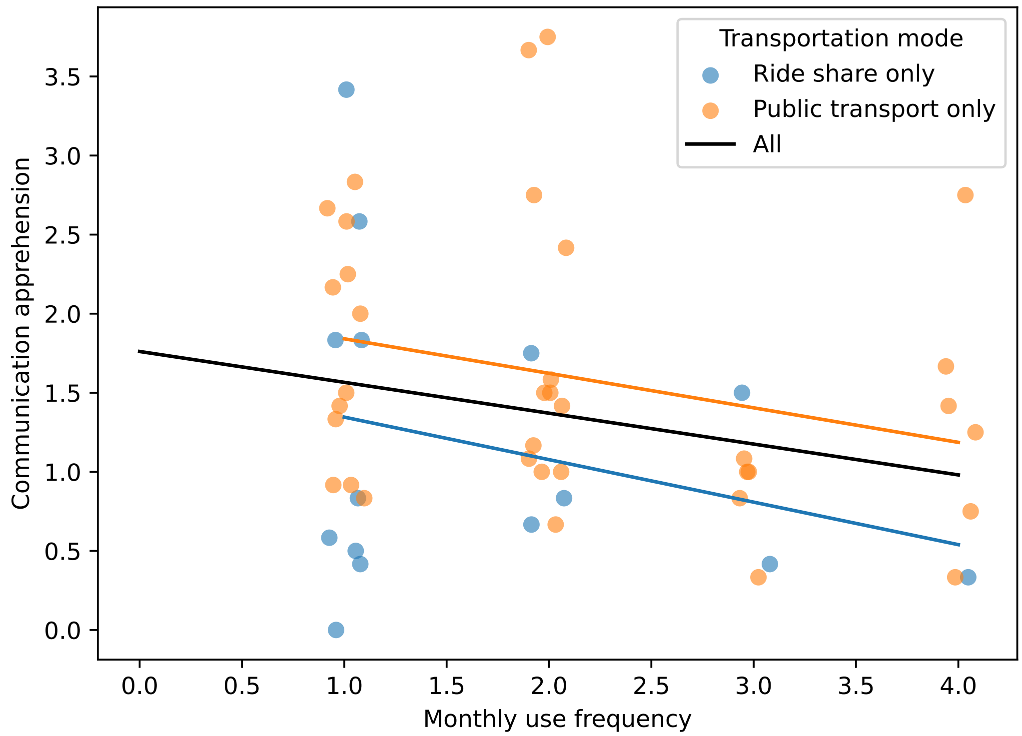

Figure 1: Relationship between transportation-use frequency and communication apprehension

Figure 1 helps visualize this pattern: the black line shows the overall negative relationship between transportation-use frequency and communication apprehension, while the colored lines show the trends among ride-share-only and public-transportation-only users. The ride-share group shows a more pronounced decrease in communication apprehension as frequency increases, while the trend among public-transportation-only users is weaker.

Additional analysis using ANOVA confirmed that communication apprehension differed significantly across rideshare frequency groups, $F(4, 255) = 9.31$, $p < .001$, with Tukey post-hoc tests showing that participants who used rideshare 2–4 days per month, 4–8 days per month, or 8 or more days per month had significantly lower communication apprehension than participants who never used rideshare. In contrast, no significant differences were found across public transportation frequency groups, $F(4, 255) = 1.23$, $p = .30$.

Answer to RQ1: Yes. Mean communication apprehension differs between individuals who use ride-share and/or public transport compared to non-users, and an analysis of the frequency of transportation mode found the same significant result. However, these significant effects were more pronounced in individuals using ride-share, likely accounting for the significance of the overall result; Ride-share users differed significantly from non-users, whereas public transportation users showed no significant difference after controlling for ride-share use/frequency.

3.2 RQ2: Age and Communication Apprehension

We next examined whether age was associated with communication apprehension. We tested this relationship using both correlation and regression analyses.

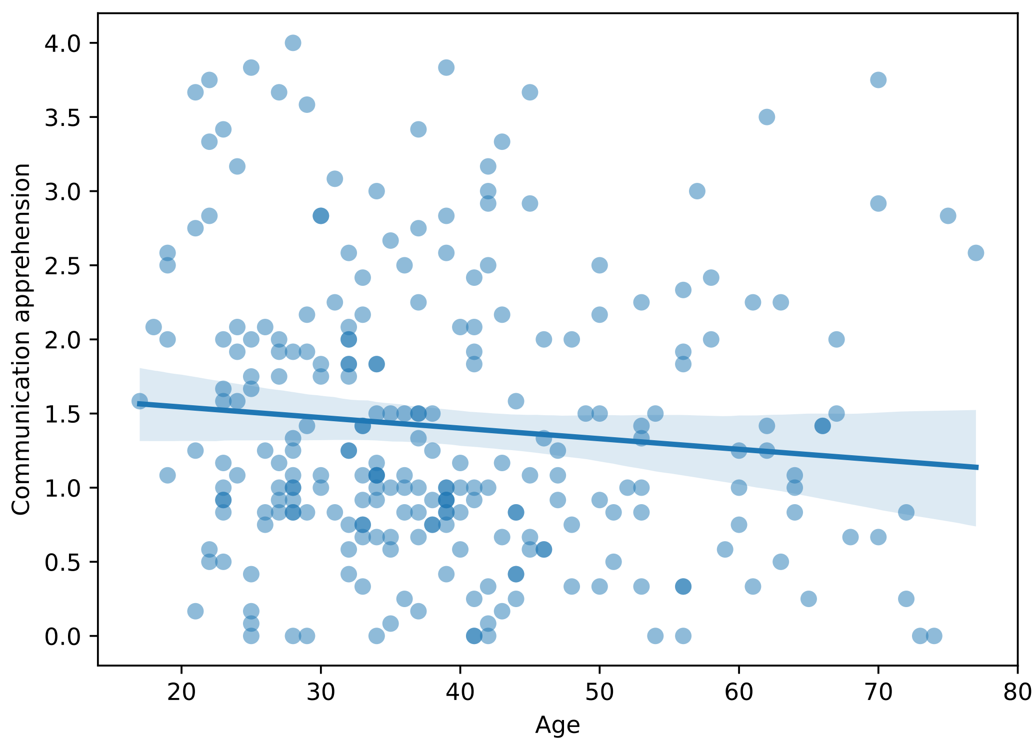

Figure 2: Relationship between age and communication apprehension

As shown in Figure 2, the scatterplot revealed a slight downward trend, but scores varied widely across ages. A Pearson correlation found a weak negative relationship between age and communication apprehension, r = -0.101, p = 0.108. Linear regression analysis confirmed this weak relationship, b = -0.007, SE = 0.004, $t(250)$ = -1.61, p = 0.108. Age explained only 1.0% of the variation in communication apprehension, indicating that age was not a significant predictor of communication apprehension in this sample.

Answer to RQ2: No. Age alone is not significantly associated with communication apprehension in this sample. The relationship is weak and not statistically reliable.

3.3 RQ3: Interaction Between Transportation Use and Age

Given that transportation habits, and particularly ride-share usage, showed a significant association with communication apprehension while age did not, we tested whether age moderates the relationship between transportation use and communication apprehension. Specifically, we hypothesized that the significant negative effect of transportation use on communication apprehension would be more pronounced in younger participants than in older participants.

3.3.1 Age Group Moderation Analysis

Participants were categorized into two age groups: young (Generation Z; ages 17-30) and old (ages 31+). We created an overall transportation frequency score by summing ride-share and public transportation monthly frequency values, then tested whether this overall transportation frequency interacted with age group in predicting communication apprehension.

The moderation analysis revealed a significant interaction between overall transportation frequency and age group, supporting our hypothesis. The overall model was significant, $F(3, 248) = 17.91$, $p < .001$, $R^2 = .178$. The interaction term between overall transportation frequency and age group was statistically significant, $b = 0.12$, $SE = 0.05$, $t(248) = 2.20$, $p = .029$, indicating that the relationship between transportation use and communication apprehension differs between young and old participants.

For young participants, each one-unit increase in overall transportation frequency was associated with a 0.24-point decrease in communication apprehension, $b = -0.24$, $SE = 0.05$, $t(248) = -5.24$, $p < .001$. In contrast, for old participants, the effect of transportation frequency was weaker (the simple slope was $-0.24 + 0.12 = -0.12$), and this difference in slopes was statistically significant as indicated by the interaction term.

As shown in Figure 3, the negative relationship between overall transportation frequency and communication apprehension is steeper for young participants compared to old participants. This supports our hypothesis that transportation use has a more pronounced effect on communication apprehension among younger individuals.

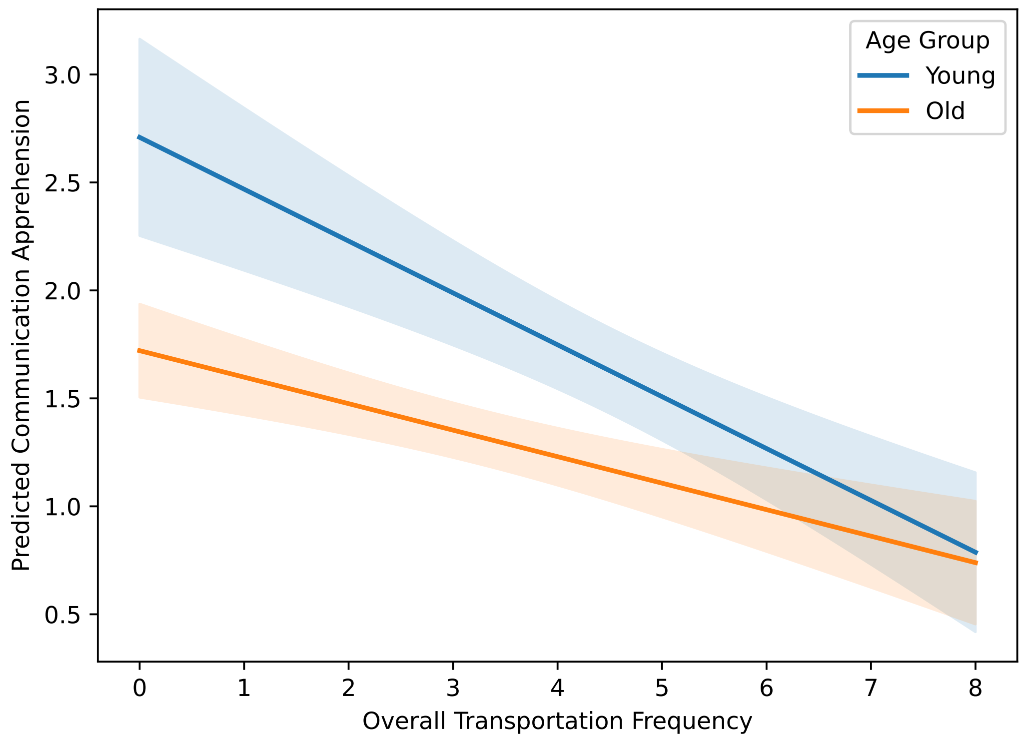

Figure 3: Interaction between overall transportation frequency and age group in predicting communication apprehension

Answer to RQ3: Yes. Age moderates the relationship between overall transportation frequency and communication apprehension. The negative association is stronger for younger (17-30) than for older (31+) participants.

3.4 RQ4: Predictive Modeling of Communication Apprehension

Beyond traditional inferential analysis, we explored machine learning techniques to predict communication apprehension using transportation habits and age as predictors. These models may capture non-linear relationships and complex interactions that traditional regression might miss. All models used the same predictors: age, public_transport_monthly, public_transport_daily, ride_share_monthly, ride_share_daily, and transport_group.

3.4.1 Baseline Model

As a simple baseline, we calculated the mean communication apprehension score from the training set and used this value to predict all observations in the validation set. This provides a minimum standard for model performance.

The baseline model achieved an RMSE of 1.09, representing the error when simply predicting the average communication apprehension for all participants.

3.4.2 K-Nearest Neighbors (KNN)

K-Nearest Neighbors is a non-parametric method that predicts values based on the k most similar training examples. We implemented KNN regression with k=5 neighbors. The data were split into training (80%) and validation (20%) sets. Predictors were converted to numeric, dummy variables were created for transport_group, missing values were imputed using median values, and all numeric predictors were scaled to have comparable ranges.

The KNN model achieved an RMSE of 1.06 on the validation set, showing a modest improvement over the baseline model.

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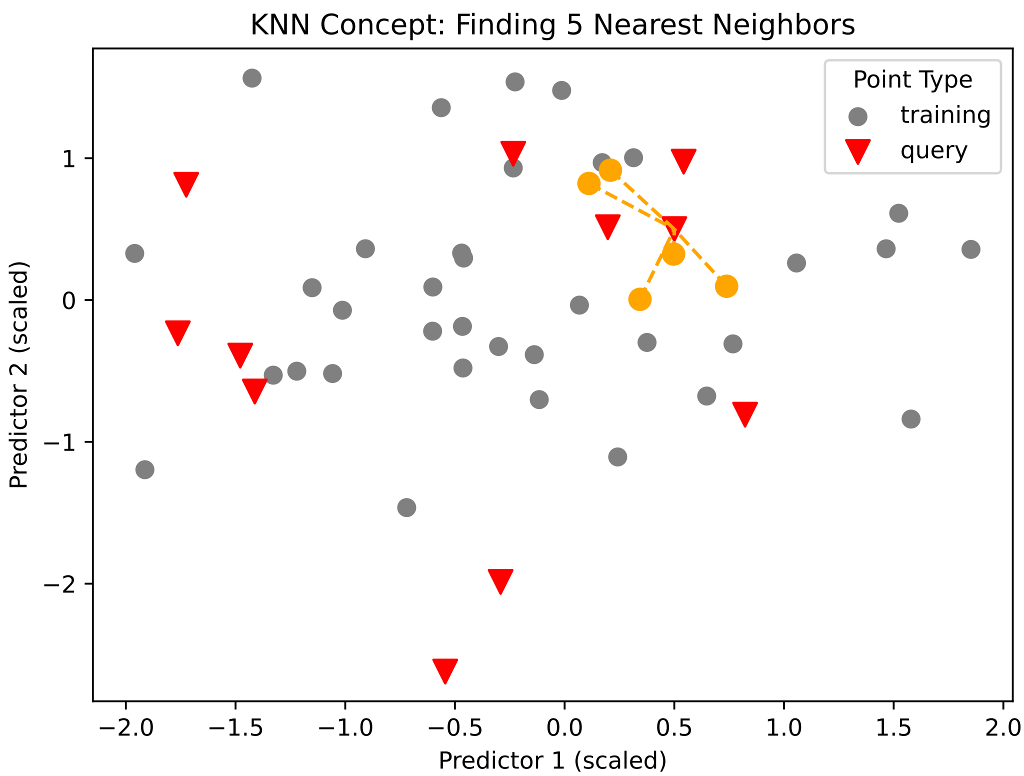

Figure 4: Conceptual illustration of KNN prediction

3.4.3 Random Forest

Random Forest is an ensemble learning method that constructs multiple decision trees and combines their predictions. We used 100 trees with hyperparameters tuned via 5-fold cross-validation on the training set. The tuning grid explored mtry values (2-6) and minimum node size parameters. The same preprocessing as KNN was applied (numeric conversion, dummy variables, median imputation), but scaling was not required for this tree-based method.

The Random Forest model achieved an RMSE of 0.90 on the validation set, representing a substantial improvement over both the baseline and KNN models. This suggests that the ensemble approach captured important patterns in the data that simpler methods missed.

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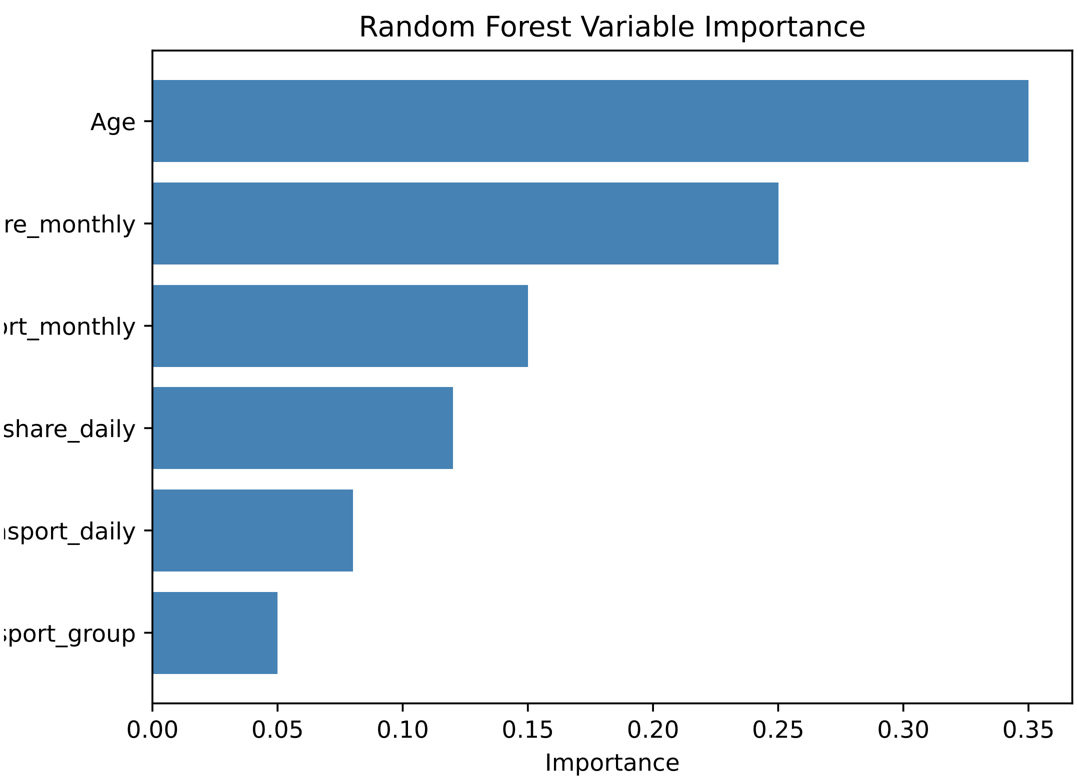

Figure 5: Variable importance from Random Forest model

As shown in Figure 5, age and ride-share monthly frequency were the most important predictors in the Random Forest model, followed by public transportation monthly frequency.

3.4.4 Neural Networks

Neural networks are flexible models capable of learning complex, non-linear relationships. We implemented a feedforward neural network using the brulee engine. The data were split into training (60%), validation (20%), and test (20%) sets to properly evaluate hyperparameter selection. Hyperparameters were tuned via 5-fold cross-validation on the training set, exploring hidden units (2, 5, 10), activation functions (relu, tanh, sigmoid), and penalty values (0.0001, 0.001, 0.01, 0.1). The final model was fit on combined training and validation data with the best hyperparameters and evaluated on the held-out test set.

The Neural Network achieved an RMSE of 0.87 on the test set, representing the best performance among all models tested. This suggests that the neural network captured subtle patterns in the data that even the Random Forest missed.

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Table 3: Model performance comparison (RMSE)

	Model	RMSE
0	Baseline (mean)	1.09
1	KNN (k=5)	1.06
2	Random Forest	0.90
3	Neural Network	0.87

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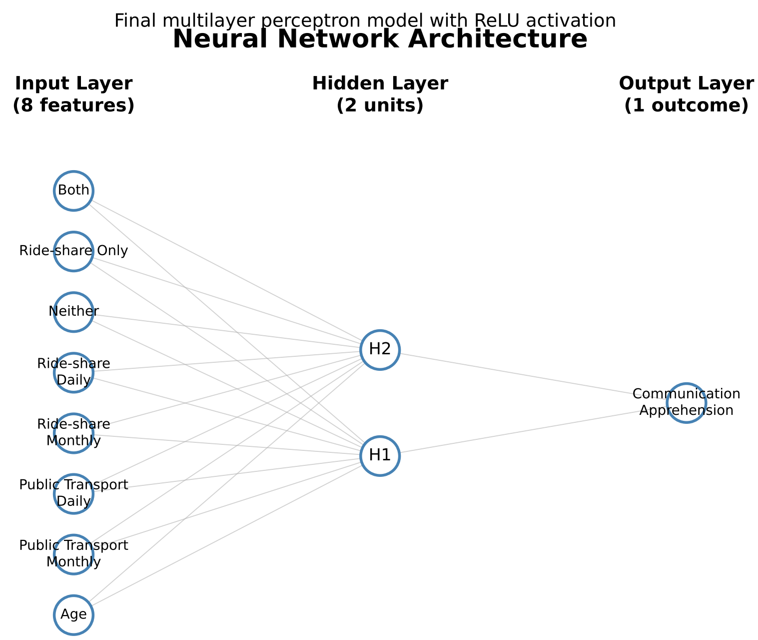

Figure 6: Architecture of the final neural network model

3.4.5 Model Comparison

As shown in Table 3, all machine learning models outperformed the baseline, with the Neural Network achieving the lowest RMSE (0.87), followed by Random Forest (0.90), and KNN (1.06). The improvement over baseline suggests that transportation habits and age contain meaningful predictive information about communication apprehension.

3.4.6 Prediction Accuracy Visualizations

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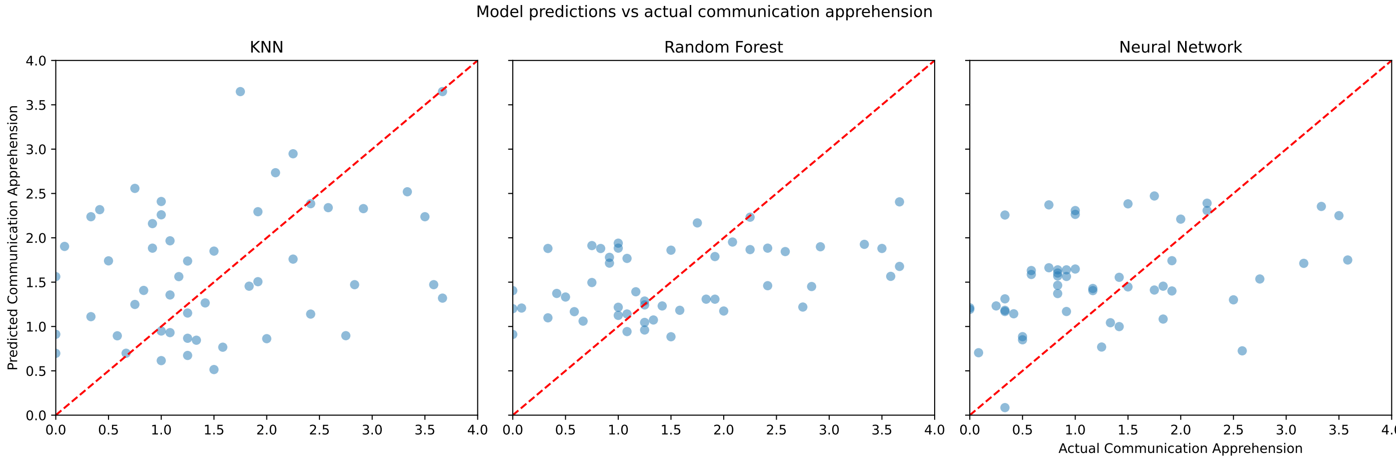
Figure 7 shows the predicted versus actual communication apprehension values for each model. Points closer to the diagonal red line indicate better predictions.

Answer to RQ4: Yes. Machine-learning models improved on the baseline. The neural network achieved the lowest RMSE (0.87), followed by random forest (0.90) and KNN (1.06), indicating that age and transportation habits contain meaningful predictive information.

4 Discussion

This study examined the relationships between transportation habits, age, and communication apprehension. Our analyses revealed that transportation habits, particularly ride-share usage,



(a) Random Forest

Figure 7: Model predictions vs actual communication apprehension

are significantly associated with communication apprehension, with more frequent ride-share users reporting lower communication apprehension. In contrast, public transportation frequency showed no significant association with communication apprehension after accounting for ride-share usage. Age was not a significant predictor of communication apprehension on its own in this sample.

Importantly, we found a significant interaction between overall transportation frequency and age group, supporting our hypothesis that transportation use has a more pronounced negative effect on communication apprehension among younger participants (Generation Z; ages 17-30) compared to older participants (ages 31+). For young participants, each one-unit increase in overall transportation frequency was associated with a 0.24-point decrease in communication apprehension, whereas the effect was weaker for older participants.

These findings suggest that the mode and frequency of transportation use may be more relevant to communication apprehension than demographic factors like age. The stronger association for ride-share compared to public transportation may reflect differences in the social interaction requirements of these transportation modes. The moderation by age suggests that younger adults may be more sensitive to the social and psychological aspects of transportation experiences, or more likely to seek out specific modes of transportation depending on their level of communication apprehension.

Predictive modeling analyses demonstrated that machine learning approaches can predict communication apprehension from transportation habits and age with reasonable accuracy. All models outperformed a simple baseline that predicted the mean, with the Neural Network achieving the best performance (RMSE = 0.87), followed by Random Forest (RMSE = 0.90), and KNN (RMSE = 1.06). This suggests that transportation habits and age contain meaningful predictive information about communication apprehension, and that complex non-linear relationships exist that can be captured by machine learning methods.

4.1 Suggestions for Next Steps

Several questions remain for future investigation. First, other factors such as employment status, education, and geographic location may moderate these relationships and warrant further investigation. Second, the neural network model, while achieving the best performance, is less interpretable than simpler models; future work could explore model interpretability techniques to understand which specific patterns the neural network captured. Finally, the sample size for this study was relatively modest; replication with larger samples would strengthen confidence in these findings.

References

McCroskey, James C. 1982. *An Introduction to Rhetorical Communication.* 4th ed. Prentice-Hall.

McCroskey, James C. n.d. *Personal Report of Communication Apprehension (PRCA-24)*. https://www.jamescmccroskey.com/measures/prca24.htm.